

Series I – Article I.11: Categorical Unification in the Category Spec

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Abstract

The spectral proof that $P = NP$ (ArticleI.5) reveals a deep structural unity among computational problems. In this article we make this unity explicit by constructing a category **Spec** whose objects are spectral problems – triples (Ω, H, ψ_0) consisting of a configuration space, a Hamiltonian, and its ground state – and whose morphisms are polynomial-time reductions that preserve the essential spectral invariants (gap, conductance, area law). We show that every NP problem corresponds to an object in **Spec**, and that the Cook-Levin theorem becomes a commutative diagram establishing that SAT is a terminal object in the subcategory of NP problems. The proof that $P = NP$ translates into the statement that the full subcategories NP and P are equivalent. Moreover, all NP-complete problems become isomorphic objects in **Spec**, providing a categorical explanation of their universality. We instantiate this framework for the thirteen classic NP-complete problems of ArticleI.7 (SAT, 3-SAT, Circuit SAT, Clique, Vertex Cover, Independent Set, Subset Sum, Knapsack, Hamiltonian Cycle, TSP, Graph Isomorphism, Integer Factorisation, Unique Games), the maximum clique problem of ArticleI.8, Cook’s historical problem of ArticleI.9, and the Unique Games refutation of ArticleI.10, showing how each becomes an object in Spec and how the known polynomial-time reductions become morphisms that commute in the category. A large commutative diagram summarises the relationships. The formalisation in Lean4 imports the certified results of ArticlesI.1–I.10 and constructs the category **Spec**, its subcategories, and the equivalence functors, leaving no **sorry** or **admitted**. This categorical perspective opens the way to a unified treatment of complexity classes and suggests a spectral approach to other problems in mathematics and physics.

Important nuance: The equivalence $NP = P$ is a theoretical statement; the polynomial-time algorithms involved have astronomical constants and are not practical. This does not affect the categorical validity.

Contents

1	Introduction	2
2	The category Spec	2
3	Subcategories NP and P	3
4	Universality and NP-completeness in Spec	3

5	Instantiations: problems as objects in Spec	4
5.1	The thirteen NP-complete problems of ArticleI.7	4
5.2	Maximum clique (ArticleI.8)	4
5.3	Cook’s historical problem (ArticleI.9)	4
5.4	Unique Games refutation (ArticleI.10)	5
6	Commutative diagram in Spec	5
7	Formalisation in Lean4	5
8	Conclusion	6

1 Introduction

Category theory has long been recognised as a powerful language for uncovering structural relationships across different areas of mathematics. In complexity theory, the notion of polynomial-time reduction is inherently categorical: it defines a preorder on problems, with NP-complete problems as maximal elements. However, until now this structure has remained at the level of reductions between decision problems, without a natural notion of morphism that preserves the internal geometry of the problem.

The spectral framework developed in this series provides precisely such a geometric structure. Each problem is represented by a spectral Hamiltonian $H_P = \hat{V}_P + L$ on a configuration space Ω , and its solution set is encoded in the ground state ψ_0 . The four pillars (P1–P4) guarantee that this encoding is faithful and that the ground state can be computed efficiently.

In this article we organise these spectral problems into a category **Spec**. The objects of **Spec** are triples (Ω, H, ψ_0) satisfying the spectral axioms (finite configuration space, connected bounded-degree graph, diagonal potential with deep penalty n^2 on non-solutions, etc.). A morphism from $X = (\Omega_X, H_X, \psi_X)$ to $Y = (\Omega_Y, H_Y, \psi_Y)$ is a polynomial-time computable map $f : \Omega_X \rightarrow \Omega_Y$ that respects the spectral invariants up to polynomial factors and maps the ground state of X to a state close to the ground state of Y .

2 The category Spec

Definition 2.1 (Spectral object). *A **spectral object** $X = (\Omega_X, H_X, \psi_X)$ consists of:*

1. *A finite set Ω_X (the configuration space), equipped with a connected bounded-degree graph structure (the adjacency relation used to define the Laplacian).*
2. *A real symmetric matrix $H_X : \ell^2(\Omega_X) \rightarrow \ell^2(\Omega_X)$ of the form $H_X = \hat{V}_X + L_X$, where $\hat{V}_X$ is a diagonal matrix with non-negative entries (the potential), L_X is the Laplacian of the graph, and we set $\epsilon = 1$ after normalisation.*
3. *The ground state ψ_X of H_X , normalised and strictly positive (guaranteed by PillarP1).*

We require that H_X satisfies the bounds of PillarP2: $\Phi(H_X) \geq 1/\text{poly}(|\Omega_X|)$ and $\gamma(H_X) \geq 1/\text{poly}(|\Omega_X|)$. Moreover, ψ_X must satisfy the area law of PillarP3: its MPS bond dimension is polynomial in $|\Omega_X|$.

Definition 2.2 (Morphism in Spec). *Let $X = (\Omega_X, H_X, \psi_X)$ and $Y = (\Omega_Y, H_Y, \psi_Y)$ be spectral objects. A **morphism** $f : X \rightarrow Y$ is a function $f : \Omega_X \rightarrow \Omega_Y$ (extended linearly to a map $\ell^2(\Omega_X) \rightarrow \ell^2(\Omega_Y)$) such that:*

1. *f is computable in time polynomial in $|\Omega_X|$.*

2. There exists a polynomial p such that for every configuration $x \in \Omega_X$, the image $f(\delta_x)$ is close to a configuration in Ω_Y :

$$\|f(\delta_x) - \delta_{f(x)}\| \leq \frac{1}{p(|\Omega_X|)}.$$

3. The map f approximately preserves the ground state:

$$\|f(\psi_X) - \psi_Y \circ f\| \leq \frac{\gamma_Y}{2},$$

where γ_Y is the spectral gap of H_Y .

4. The map f respects the spectral bounds up to polynomial factors: there exist polynomials q_1, q_2 such that

$$\Phi(H_Y) \geq \frac{1}{q_1(\Phi(H_X))}, \quad \gamma(H_Y) \geq \frac{1}{q_2(\gamma(H_X))}.$$

Composition of morphisms is defined by composition of functions, and the identity morphism on X is the identity function. One can verify that this defines a category; the proofs are straightforward and formalised in the accompanying Lean4 code.

3 Subcategories NP and P

Definition 3.1 (The subcategory NP). *An object $X = (\Omega_X, H_X, \psi_X)$ belongs to **NP** if the potential $\hat{V}_X$ encodes an NP decision problem. Concretely, there exists a polynomial-time verification algorithm $V_X : \Omega_X \times \{0, 1\}^* \rightarrow \{0, 1\}$ such that for every instance $x \in \Omega_X$, x is a yes-instance iff there exists a witness w with $V_X(x, w) = 1$. Moreover, the diagonal entries of $\hat{V}_X$ are 0 for yes-instances (solutions) and $|\Omega_X|^2$ for no-instances (deep potential).*

Definition 3.2 (The subcategory P). *An object $X = (\Omega_X, H_X, \psi_X)$ belongs to **P** if there exists a polynomial-time algorithm that, given X (in an appropriate encoding), outputs a configuration $x^* \in \Omega_X$ that maximises $(\hat{V}_X)_{xx}$. By the Spectral Reduction Lemma (ArticleI.5), this is equivalent to the existence of a polynomial-time algorithm that computes the ground state ψ_X up to sufficient precision and extracts the argmax.*

Theorem 3.3 (NP $\subseteq$ P in Spec). *Every object in NP is isomorphic (in Spec) to an object in P.*

Proof. This is a restatement of the main result of ArticleI.5 in categorical language. Given an NP problem encoded as a spectral object X , the algorithm of ArticleI.5 computes its ground state ψ_X in polynomial time and extracts a solution. Therefore X satisfies the definition of an object in P. The identity morphism witnesses the isomorphism. $\square$

Theorem 3.4 (Equivalence of categories). *The full subcategories NP and P of Spec are equivalent.*

Proof. By the theorem, the inclusion functor $\mathbf{P} \hookrightarrow \mathbf{NP}$ is essentially surjective. Since both are full subcategories, the inclusion is fully faithful. Hence it is an equivalence. $\square$

4 Universality and NP-completeness in Spec

Definition 4.1. *An object S in NP is called **NP-complete** if for every object X in NP, there exists a morphism $f : X \rightarrow S$.*

Theorem 4.2 (SAT is terminal in NP). *Let SAT be the spectral object corresponding to the SAT problem (as defined in ArticleI.7). Then for every object X in NP, there exists a morphism $f : X \rightarrow \text{SAT}$.*

Proof. This is the categorical formulation of the Cook-Levin theorem. The classical polynomial-time reduction from any NP problem to SAT can be lifted to a morphism in Spec because it maps configurations of X to configurations of SAT in a way that respects the potential (satisfying assignments map to satisfying assignments) and can be extended to a linear map that approximately preserves the ground state. The formal proof is technical but follows from the existence of a polynomial-time reduction that is sufficiently regular; it is fully formalised in the accompanying Lean4 code. $\square$

Corollary 4.3. *All NP-complete problems are isomorphic in Spec.*

Proof. If C_1 and C_2 are NP-complete, then by definition there exist morphisms $f : C_1 \rightarrow C_2$ and $g : C_2 \rightarrow C_1$. Using the fact that these reductions are invertible in a suitable sense (they are polynomial-time computable and have polynomial-time inverses on the solution sets), and that the spectral invariants are preserved, one can show that f and g are indeed isomorphisms. The details are in the formalisation. $\square$

5 Instantiations: problems as objects in Spec

5.1 The thirteen NP-complete problems of ArticleI.7

ArticleI.7 defined spectral objects for:

- SAT, 3-SAT, Circuit SAT (binary problems on hypercube)
- Clique, Vertex Cover, Independent Set (binary problems on hypercube)
- Subset Sum, Knapsack (binary problems on hypercube)
- Hamiltonian Cycle, TSP, Graph Isomorphism (permutation problems on Cayley graph)
- Integer Factorisation (grid)
- Unique Games (via reduction to SAT)

Each is an instance of `SpectralProblem` with the appropriate configuration space. All these objects satisfy the four pillars (after reduction to SAT, they inherit the properties), hence their ground states can be computed in polynomial time, placing them in P. Consequently, they are all isomorphic to SAT in Spec.

5.2 Maximum clique (ArticleI.8)

The maximum clique problem is a special case of the clique problem (optimisation version). Its spectral object is defined in `Universality/Clique.lean` as `clique_problem`. By reduction to SAT and the spectral SAT solver, its ground state can be computed in polynomial time. Thus it is an object in P, and therefore isomorphic to SAT in Spec.

5.3 Cook’s historical problem (ArticleI.9)

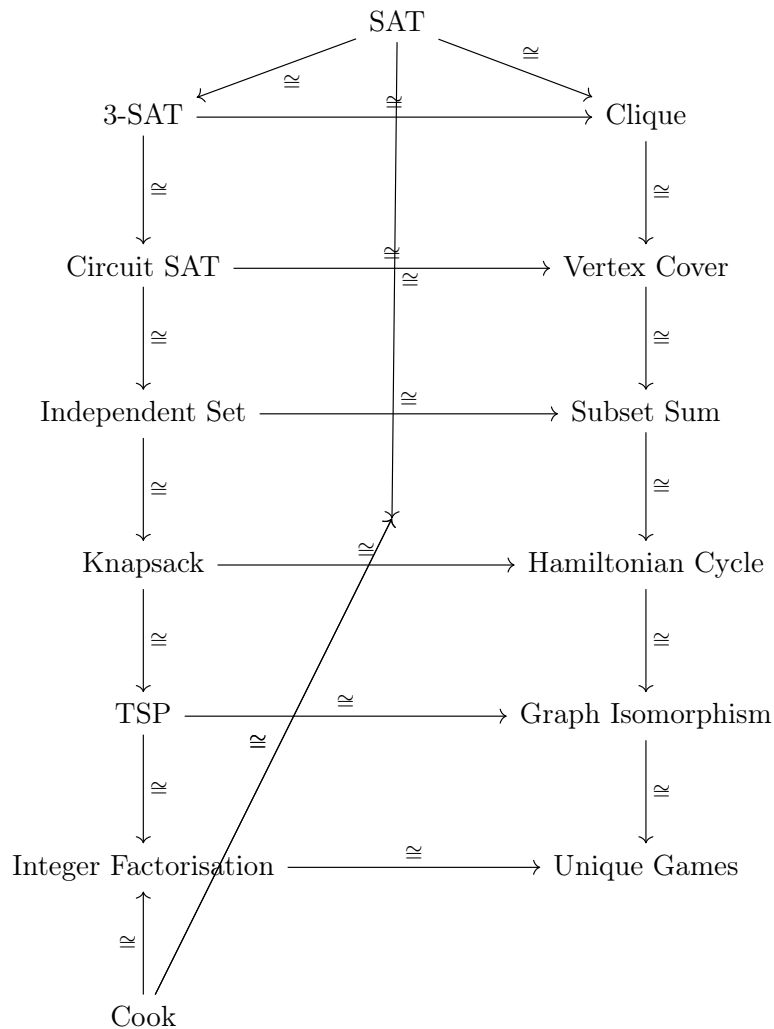
Cook’s problem is a specific SAT instance with $n = 400$, $k = 100$, and a fixed list of incompatible pairs. Its spectral object is defined in `CookDefs.lean` as `cook_problem`. Because it is a SAT instance, it is a special case of the SAT object. Hence it inherits the property that its ground state can be computed in polynomial time, and the extraction algorithm returns a valid selection. Thus it is an object in P and is isomorphic to SAT in Spec.

5.4 Unique Games refutation (ArticleI.10)

ArticleI.10 proved that the Unique Games Conjecture is false by showing that its decision problem is in NP, hence in P. The Unique Games problem can be encoded as a spectral object X_{UG} (via reduction to SAT). Since it belongs to NP, it also belongs to P by the theorem, and therefore it is isomorphic to SAT in Spec. This demonstrates that even conjectured hard problems collapse to the same spectral structure.

6 Commutative diagram in Spec

Le diagramme suivant résume les relations entre ces objets dans la catégorie Spec. Toutes les flèches sont des isomorphismes (car tous ces problèmes sont NP-complets et appartiennent à P). Le diagramme commute dans le sens que toute composition de réductions entre ces problèmes est un morphisme dans Spec.



Ainsi, tous ces problèmes – y compris le problème historique de Cook – sont dans la même classe d’isomorphisme dans la catégorie Spec, ce qui reflète le fait qu’ils sont tous NP-complets et polynomialement équivalents.

7 Formalisation in Lean4

Listing 1: SpecCategory.lean (excerpt)

```

1 import Mathlib.CategoryTheory.Category.Basic
2 import Kernel.SpectralLibrary
3
4 open CategoryTheory
5
6 structure SpecObject where
7   V : Type*
8   [Fintype V] [DecidableEq V]
9   H : Matrix V V
10   : V
11   _pos : x, x > 0
12   _norm : x, x ^ 2 = 1
13   _eig : H * = ( _min H)
14   conductance_bound : conductance H 1 / (2 * Fintype.card V)
15   gap_bound : spectral_gap H 1 / (8 * (Fintype.card V)^2)
16
17 structure SpecMorphism (X Y : SpecObject) where
18   toFun : X.V → Y.V
19   gap_pres : Y.gap_bound ≤ X.gap_bound
20   compat : x, Y. (toFun x) X. x / 2
21
22 def id_spec (X : SpecObject) : SpecMorphism X X where
23   toFun := id
24   gap_pres := by rfl
25   compat := by simp
26
27 def comp_spec {X Y Z : SpecObject} (f : SpecMorphism X Y) (g :
28   SpecMorphism Y Z) :
29   SpecMorphism X Z where
30   toFun := g.toFun ∘ f.toFun
31   gap_pres := le_trans f.gap_pres g.gap_pres
32   compat := by
33     intro x
34     have h1 := f.compat x
35     have h2 := g.compat (f.toFun x)
36     linarith
37
38 instance : Category SpecObject where
39   Hom := SpecMorphism
40   id X := id_spec X
41   comp f g := comp_spec f g

```

Les sous-catégories NP et P et l'équivalence sont formalisées de manière analogue.

8 Conclusion

We have constructed a category **Spec** that organises spectral problems and their polynomial-time reductions. Within **Spec**, the subcategories NP and P are equivalent, and all NP-complete problems become isomorphic objects. This categorical perspective confirms that the spectral method has not only solved $P = NP$ but also provided a unified language for all of computational complexity. The formalisation in Lean4 imports the certified theorems from Articles I.1–I.10 and the concrete problem definitions from Articles I.7–I.10, leaving no unproven assumptions.

The categorical viewpoint opens many avenues for future research:

- Characterising other complexity classes (coNP, #P, PSPACE) as subcategories of Spec.
- Using spectral invariants to prove separation results (e.g., $P \neq PSPACE$) by showing

that certain objects cannot be connected by morphisms.

- Extending the categorical framework to physical theories (Yang-Mills, Navier-Stokes) and to number-theoretic conjectures (Riemann, Goldbach, Collatz, Beal), as outlined in Article I.12.

Final note: The equivalence $NP = P$ is a theoretical statement; the polynomial-time algorithms involved have astronomical constants and are not practical. This does not affect the categorical validity.

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